

Project 1

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Before starting this lab, I reviewed both codes of ethics. The goal of this lab is to understand password security, test password-cracking tools, learn how password managers improve security, and demonstrate how keyloggers work so proper defenses can be applied.

Part 1 password cracking tools

Tool 1 -john the ripper

```
kmandal@ict251:~/Downloads$ john --wordlist=/home/kmandal/Downloads/rockyou.txt /home/kmandal/Desktop/hash.txt
No password hashes loaded (see FAQ)
kmandal@ict251:~/Downloads$ john --version
Unknown option: "--version"
kmandal@ict251:~/Downloads$ rm /home/kmandal/Desktop/hash.txt
kmandal@ict251:~/Downloads$ openssl passwd -1 12345678 >> /home/kmandal/Desktop/hash.txt
kmandal@ict251:~/Downloads$ openssl passwd -1 password >> /home/kmandal/Desktop/hash.txt
kmandal@ict251:~/Downloads$ openssl passwd -1 abc123 >> /home/kmandal/Desktop/hash.txt
kmandal@ict251:~/Downloads$ openssl passwd -1 Letmein >> /home/kmandal/Desktop/hash.txt
kmandal@ict251:~/Downloads$ cat /home/kmandal/Desktop/hash.txt
$1$QM3Qe5j0$Qts8.jvAzjk7cfqq/chmW/
$1$I5JgJae$kLcnQKo6/oxipZVbtAfNt1
$1$S6c7LXpz$soZP21VWGjAoX61/tTkmwM1
$1$4r5LqAz3$PtGg9uxdwgn91/aqL7sS01
kmandal@ict251:~/Downloads$ john --wordlist=/home/kmandal/Download/rockyou.txt /home/kmandal/Desktop/hash.txt
Loaded 4 password hashes with 4 different salts (md5crypt [MD5 32/64 X2])
Will run 4 OpenMP threads
fopen: /home/kmandal/Download/rockyou.txt: No such file or directory
kmandal@ict251:~/Downloads$ john --wordlist=/home/kmandal/Downloads/rockyou.txt /home/kmandal/Desktop/hash.txt
Loaded 4 password hashes with 4 different salts (md5crypt [MD5 32/64 X2])
Will run 4 OpenMP threads
fopen: =/home/kmandal/Downloads/rockyou.txt: No such file or directory
kmandal@ict251:~/Downloads$ john --wordlist=/home/kmandal/Downloads/rockyou.txt /home/kmandal/Desktop/hash.txt
Loaded 4 password hashes with 4 different salts (md5crypt [MD5 32/64 X2])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
abc123          (?)
12345678        (?)
password        (?)
Letmein         (?)
4g 0:00:00:01 100% 2.030g/s 64454p/s 66014c/s 66014C/s balloo..321984
Use the "--show" option to display all of the cracked passwords reliably
Session completed
kmandal@ict251:~/Downloads$
```

John the Ripper was installed and used with the rockyou.txt wordlist to crack password hashes. It was able to crack all of the passwords pretty quickly using a dictionary attack.

Tool 2 -Hashcat

```
Watchdog: Hardware monitoring interface not found on your system.
Watchdog: Temperature abort trigger disabled.

Host memory required for this attack: 1 MB

Dictionary cache built:
* Filename..: /home/kmandal/Downloads/rockyou.txt
* Passwords.: 14344391
* Bytes.....: 139921497
* Keyspace...: 14344384
* Runtime....: 0 secs

$1$S6c7LXpz$0zP21VWGjAoX61/tTkmwM1:abc123
$1$lI5JgJae$klCnQKo6/oxipZVbtAfNt1:password
$1$QM3Qe5j0$Qts8.jVAzjk7cfqq/chmW/:12345678
$1$4r5LqAz3$PtGg9uxdwgn91/aqL7sS01:Letmein

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 500 (md5crypt, MD5 (Unix), Cisco-IOS $1$ (MD5))
Hash.Target.....: /home/kmandal/Desktop/hash.txt
Time.Started.....: Tue Feb 17 03:35:00 2026 (5 secs)
Time.Estimated...: Tue Feb 17 03:35:05 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (/home/kmandal/Downloads/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 27346 H/s (8.43ms) @ Accel:64 Loops:1000 Thr:1 Vec:4
Recovered.....: 4/4 (100.00%) Digests (total), 4/4 (100.00%) Digests (new), 4/4 (100.00%) Salts
Progress.....: 506624/57377536 (0.88%)
Rejected.....: 0/506624 (0.00%)
Restore.Point....: 126464/14344384 (0.88%)
Restore.Sub.#1...: Salt:2 Amplifier:0-1 Iteration:0-1000
Candidate.Engine.: Device Generator
Candidates.#1....: MIKAELA -> 890820

Started: Tue Feb 17 03:34:40 2026
Stopped: Tue Feb 17 03:35:05 2026
kmandal@ict251:~/Desktop$ hashcat --show -m 500 ~/Desktop/hash.txt
$1$QM3Qe5j0$Qts8.jVAzjk7cfqq/chmW/:12345678
$1$lI5JgJae$klCnQKo6/oxipZVbtAfNt1:password
$1$S6c7LXpz$0zP21VWGjAoX61/tTkmwM1:abc123
$1$4r5LqAz3$PtGg9uxdwgn91/aqL7sS01:Letmein
kmandal@ict251:~/Desktop$
```

Hashcat successfully cracked all of the passwords and showed detailed stats such as hash rate, progress, and time.

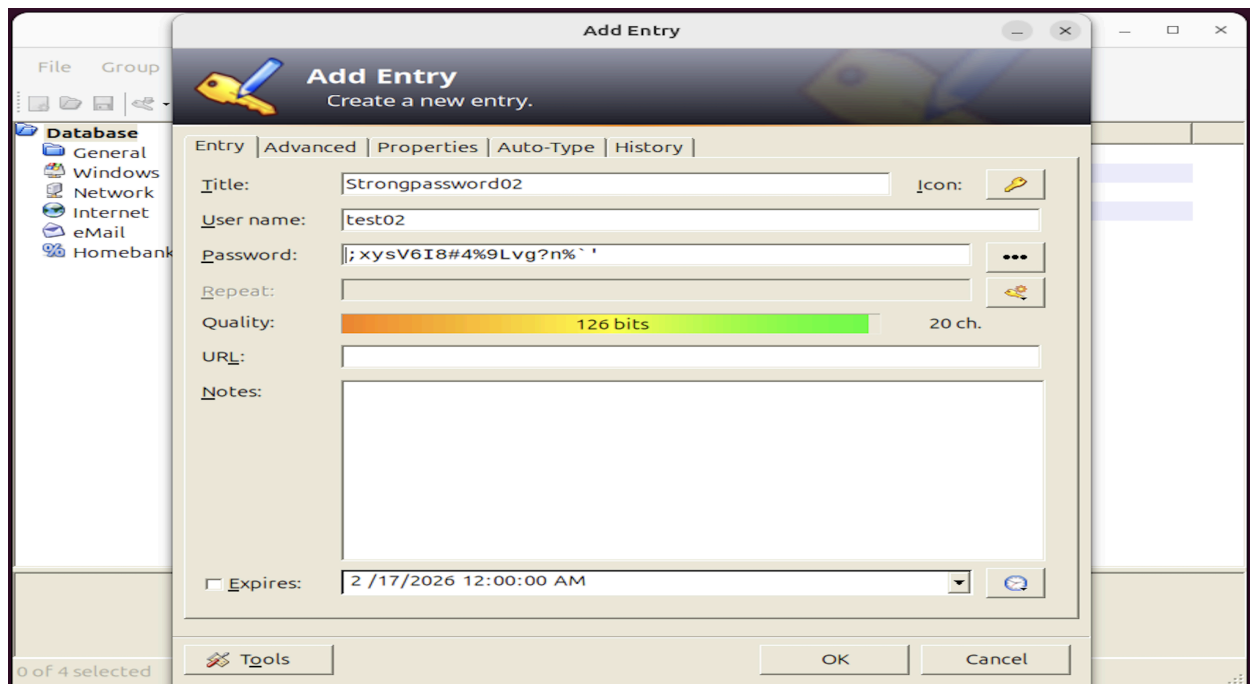
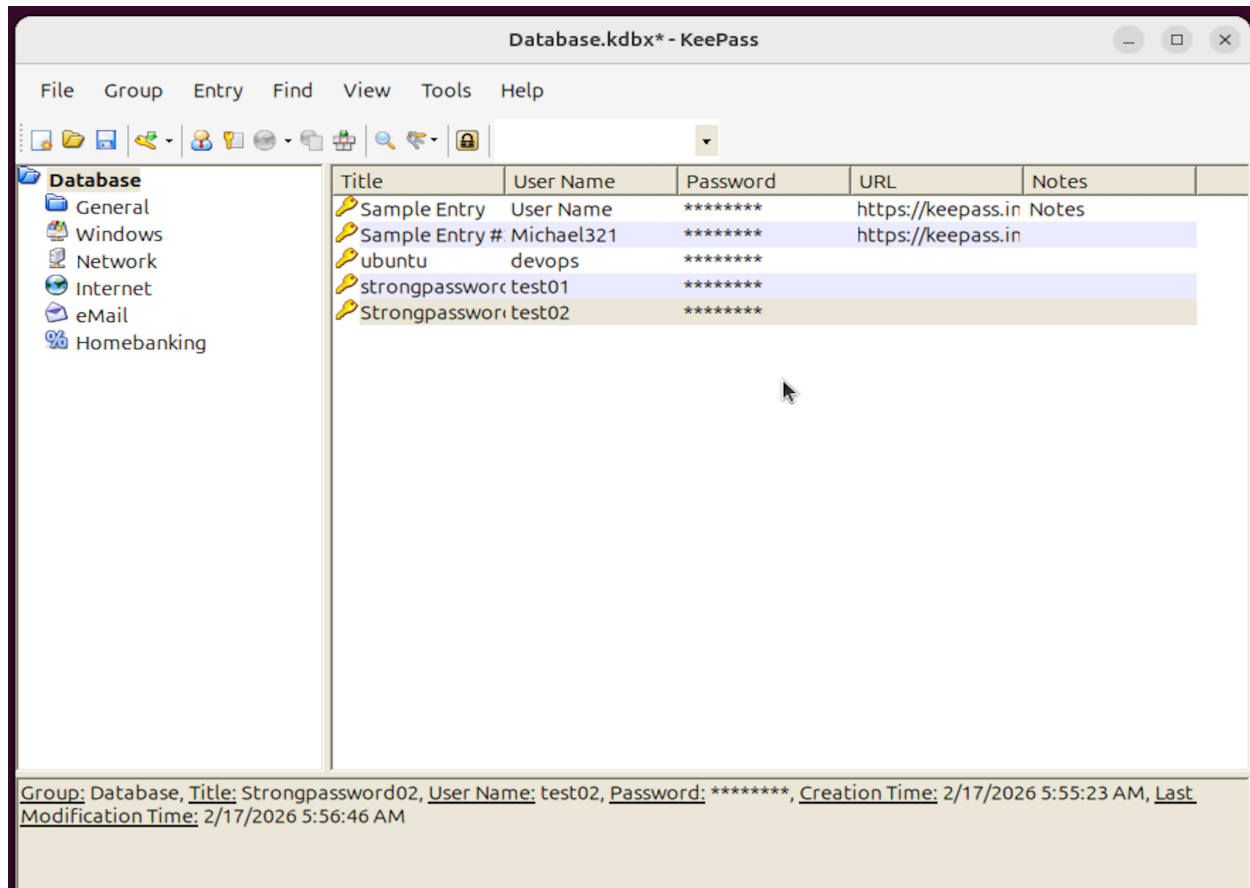
Which Tool Was Faster?

-Hashcat was the fastest tool in this experiment. It processed password hashes at a higher speed compared to John the Ripper. Both tools worked well, but Hashcat completed the cracking slightly faster and provided more detailed performance information.

If I were given a choice, I would use hashcat because it gives me more detail, and it shows the process. If there was an error somewhere, it would show that, making it easier

Part 2: Password manager

Tool- KeePass



The password manager was easy to use and generated strong passwords. The passwords are a bit long and hard to memorize, but it would be really hard for a password-cracking tool to crack the generated passwords.

Part 3 KeyLogger

Tool- Keylogger

```
kmandal@ict251:~/Desktop$ script keylog.txt
Script started, output log file is 'keylog.txt'.
kmandal@ict251:~/Desktop$ hello this is a keylogger test
Command 'hello' not found, but can be installed with:
sudo snap install hello          # version 2.10, or
sudo apt install hello           # version 2.10-3
sudo apt install hello-traditional # version 2.10-6
See 'snap info hello' for additional versions.
kmandal@ict251:~/Desktop$ passwords123
passwords123: command not found
kmandal@ict251:~/Desktop$ qwerty
qwerty: command not found
kmandal@ict251:~/Desktop$ abc123
abc123: command not found
kmandal@ict251:~/Desktop$ all of these passwords will be captured
Command 'all' not found, did you mean:
  command 'alf' from deb seagan-apps (2.4.0+dfsg-15ubuntu1)
  command 'axl' from deb afnix (3.8.0-1)
  command 'al' from deb mono-devel (6.8.0.105+dfsg-3.5ubuntu1)
  command 'dll' from deb brickos (0.9.0.dfsg-12.2)
  command 'alc' from deb amule-utils-gui (1:2.3.3-3build1)
  command 'als' from deb atool (0.39.0-13)
  command 'al2' from deb mono-devel (6.8.0.105+dfsg-3.5ubuntu1)
  command 'ale' from deb ale (0.9.0.3-5)
  command 'alr' from deb alire (1.2.1-1.1build1)
  command 'ali' from deb mailutils-mh (1:3.16-1build1)
  command 'ali' from deb mmh (0.4-6)
  command 'ali' from deb nmh (1.8-1)
  command 'wall' from deb bsduutils (1:2.39.3-9ubuntu6.4)
Try: sudo apt install <deb name>
kmandal@ict251:~/Desktop$ password1234
password1234: command not found
kmandal@ict251:~/Desktop$
exit
Script done.
kmandal@ict251:~/Desktop$ cat keylog.txt
Script started on 2026-02-17 06:14:44+00:00 [TERM="xterm-256color" TTY="/dev/pts/3" COLUMNS="294" LINES="80"]
kmandal@ict251:~/Desktop$ hello this is a keylogger test
Command 'hello' not found, but can be installed with:
sudo snap install hello          # version 2.10, or
sudo apt install hello           # version 2.10-3
sudo apt install hello-traditional # version 2.10-6
See 'snap info hello' for additional versions.
kmandal@ict251:~/Desktop$ passwords123
passwords123: command not found
kmandal@ict251:~/Desktop$ qwerty
qwerty: command not found
kmandal@ict251:~/Desktop$ abc123
abc123: cokmandal@ict251:~/Desktop$
```

I used a simple keylogger to see how it works in the virtual machine. After starting the keylogger and typing sample text, the program recorded and displayed all keystrokes, including letters. This program showed how keyloggers can capture sensitive information such as passwords. As far as I am aware, my mac doesnt have a keylogger.